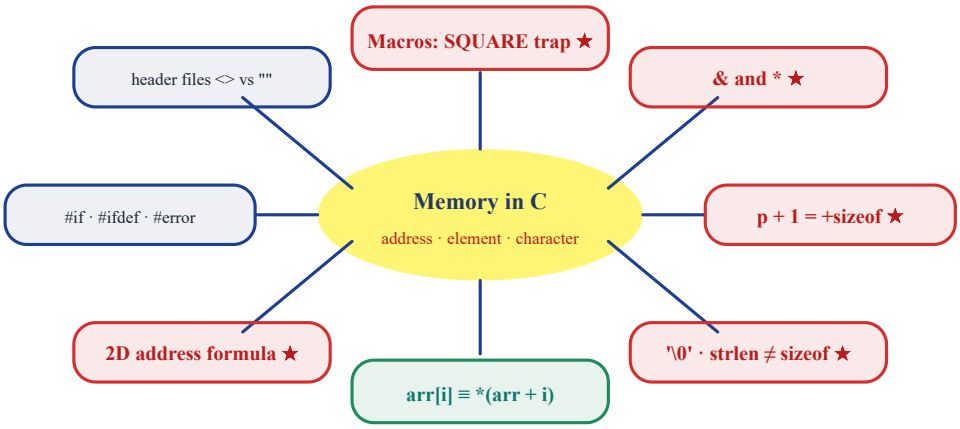
Addresses, Arrays & Strings in C

Header files · Preprocessor · Macros · Pointers · 1D/2D arrays · Strings: exam-ready in 11 sheets



↑ the whole lesson on one page: red = high-yield, green = the rule everything else hangs on

Syllabus coverage

The Structured Programming Language syllabus lists: "**Header files; Preprocessor; Pointers and arrays; Strings; Multidimensional array**". Data types, operators and control flow, and functions, scope and storage classes, were the previous lessons; structures, unions and enums, then I/O, files and `malloc`, come next.

High-yield list

① `<>` system dirs; `""` own dir first
② Preprocessor = text work before compiling
③ `SQUARE(2+3)` with `x*x` → 11
④ `&` address-of; `*` value-at
⑤ `p + 1` moves `sizeof(*p)` bytes
⑥ `arr[i] ≡ *(arr+i) ≡ i[arr]`
⑦ `sizeof(arr)` ≠ `sizeof(p)`
⑧ C is row-major
⑨ Strings end in `'\0'` (ASCII 0)
⑩ `gets()` removed in C11 → `fgets()`

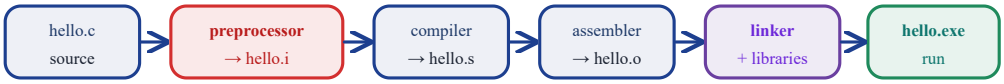
Legend: **yellow** = key term · **red circle** = must-memorise · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

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1 Header Files and the Build Pipeline

Definition: A **header file** (`.h`) holds **declarations** shared by source files: function prototypes, macros, constants and type definitions (`typedef`, `struct`). `#include` pastes the file's whole text in place of the line **before compilation**. A header holds no compiled code: the bodies of `printf`, `sqrt` … live in the **library**, joined in by the **linker**.



headers are pasted in at the preprocessor; library code joins at the linker

gcc -E stops after preprocessing, -S after compiling, -c after assembling

Header	What it gives you · names to recognise
`<stdio.h>`	standard I/O: printf, scanf, puts, fgets, getchar, fopen, FILE, EOF
`<conio.h>`	console I/O, **not standard C** (Turbo C/DOS, MSVC): clrscr, getch, getche
`<math.h>`	maths on double: sqrt, pow, fabs, ceil, floor, sin, log (gcc: link with -lm)
`<string.h>`	strings and memory blocks: strlen, strcpy, strcat, strcmp, strstr, memcpy
`<stdlib.h>`	general utilities: malloc, calloc, free, atoi, rand, exit, abs, qsort
`<ctype.h>`	character tests and case: isalpha, isdigit, isspace, toupper, tolower
`<time.h>`	date and time: time, clock, difftime, CLOCKS_PER_SEC
`<limits.h>`	integer limits: INT_MAX, INT_MIN, CHAR_BIT, LONG_MAX
`<float.h>`	floating-point limits: FLT_MAX, DBL_MAX, FLT_EPSILON, FLT_DIG
`<stddef.h>` · `<stdbool.h>`	NULL, size_t, ptrdiff_t · bool, true, false (C99)

```
/* myheader.h */
#ifndef MYHEADER_H     /* include guard */
#define MYHEADER_H
int add(int a, int b); /* prototype */
#define MAX 100
#endif
/* main.c */
#include <stdio.h>      /* system dirs */
#include "myheader.h"   /* own dir 1st */
```

<…> vs "…": `<stdio.h>` is searched in the standard (system) include directories only — use it for library headers. `"my.h"` is searched in the **own folder first**, then the system directories — use it for your own headers. An **include guard** stops a second copy, so no "redefinition" errors; `#pragma once` does the same (non-standard, widely supported).

Memory trick: **"Angle brackets = the library shelf; quotes = your own desk**, checked first, then the shelf."

Exam trap: A header does **not** link its library: with `<math.h>` gcc still needs `-lm`, else "undefined reference". `conio.h`, `clrscr()`, `getch()` are not ISO C.

Worked example ✍ which header?:
`INT_MAX` → limits.h · `toupper` → ctype.h · `malloc` → stdlib.h · `fabs` → math.h but `abs` → **stdlib.h**.

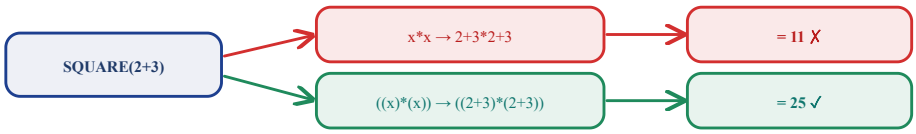
2 Preprocessor Directives and Macros

Definition: The **preprocessor** runs **before the compiler** and handles every line that starts with # (a **directive**). It does pure text work — file inclusion, **macro** substitution, conditional compilation — after comments are stripped. A directive ends at the end of its line: **no semicolon** (continue a long one with \).

Directive	Purpose
`#include`	insert a file's text
`#define`	constant or macro: `#define PI 3.1416`
`#undef`	remove a macro definition
`#ifdef` · `#ifndef`	compile if defined · not defined
`#if #elif #else #endif`	compile by a constant integer test; `defined(X)` allowed
`#error msg`	stop compilation with a message
`#pragma` · `#line`	compiler-specific order (`once`, `pack`) · reset line number

```
#define DEBUG 1
#define VERSION 2
#ifdef DEBUG
  printf("Debug mode ON\n");
#endif
#if VERSION >= 2
  newApi();
#elif VERSION == 1
  oldApi();
#else
  #error "unknown VERSION"
#endif
#undef DEBUG /* gone from here */
```

Predefined macros: `__FILE__` file name (string) · `__LINE__` line number (int) · `__DATE__` `"Oct 14 2026"` · `__TIME__` `"hh:mm:ss"` · `__STDC__` 1 on a conforming compiler · `__STDC_VERSION__` `199901L` (C99), `201112L` (C11).



a macro is replaced as text; then normal precedence applies

Point	Macro	Function
Handled	preprocessing: text	run time: a call
Type check	**none**	yes (prototype)
Arguments	re-evaluated per use	evaluated once
Speed · size	no call cost · grows	call cost · one copy
Address	none	function pointer

and ##: # stringizes: `#define STR(x) #x`, `STR(Dhaka)` → `"Dhaka"`. ## pastes tokens: `#define CAT(a,b) a##b`, `CAT(var,1)` → `var1`. Safer than a function-like macro: an `inline` function (C99).

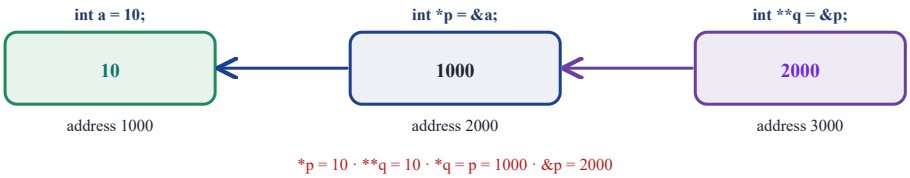
Memory trick: "**Brackets inside, brackets outside**": wrap every parameter and the whole body — `((x)*(x))`.

Exam trap: `#define PI 3.14;` pastes the `;` too. `SQUARE(i++)` increments i twice: undefined. `#define SQ (x)` with a space is object-like.

Worked example ✍ output: `#define SQ(x) (x)*(x)`: `100/SQ(5)` → `100/(5)*(5)` = 20×5 = **100**, not 4. Fully bracketed: 4.

3 Pointers: &, *, Kinds and const

Definition: A **pointer** is a variable that stores the **memory address** of another object: `type *name;`. `&` (**address-of**) gives an address; `*` (**indirection**, "value at") gives the object at an address. A pointer's own size is fixed by the machine — 4 bytes on 32-bit, **8 on 64-bit** — whatever it points to; its type decides how many bytes `*` reads and how far `+1` moves.



a pointer's box holds another box's address

```
int a = 10;
int *p;           /* declare      */
p = &a;           /* address of a */
printf("%d", *p); /* 10           */
*p = 20;          /* a is now 20  */
int **q = &p;     /* ptr to ptr   */
printf("%d", **q);/* 20           */
int *x, y;     /* x ptr, y int!   */
```

Expression	Value
`a · &a`	20 · 1000
`p · *p · &p`	1000 · 20 · 2000
`*&a` (they cancel)	20
`q · *q · **q`	2000 · 1000 · 20
`printf("%p", (void *)p)`	prints 1000

Kind	What it is	Danger and fix
NULL	`int *p = NULL;` points nowhere	dereference → crash; test `p != NULL`
void (generic)	holds any object address; `malloc` returns one	no `*v` or `v+1` without a cast: `*(int *)v`
Wild	never initialised: garbage address	initialise to `NULL` or an address
Dangling	object gone: after `free(p)`, or a local's address returned	`p = NULL;` after `free`

Declaration	`*p=5`	`p++`
`const int *p` = `int const *p`	✗	✓
`int *const p = &a;`	✓	✗
`const int *const p`	✗	✗

Look-alikes: `int *p[3]`: array of 3 pointers. `int (*p)[3]`: one pointer to an array of 3 ints. `int (*fp)(int)`: function pointer. Passing `&x` to a function: previous lesson.

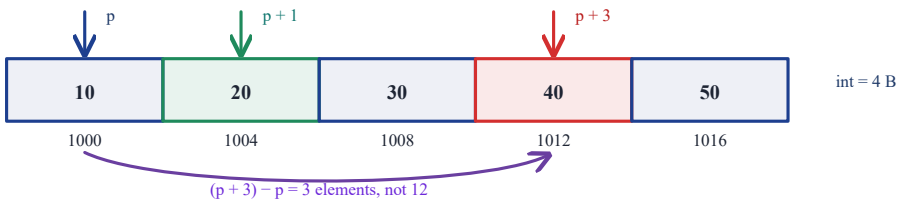
Memory trick: "**& asks where; * asks what's there.**" Read right to left: `int *const p` = "p is a const pointer to int".

Exam trap: In `int *x, y;` only **x** is a pointer: * binds to the name. A NULL or wild dereference compiles, then fails at run time.

Worked example ✍ trace: `int a = 5, *p = &a, **q = &p; **q += *p;` → both name a, so a = 5 + 5 = **10**.

4 Pointer Arithmetic

Definition: Adding an integer n to a pointer moves it **n elements**, not n bytes: **address(p + n) = address(p) + n × sizeof(*p)**. Subtracting two pointers into the same array gives the **number of elements** between them (type `ptrdiff_t`). Results are defined only inside one array, plus the position **one past the end**.



int a[5] = {10,20,30,40,50}; int *p = a; — each step jumps 4 bytes

Pointer type	sizeof(*p)	p+1 from 1000
`char *`	1	1001
`short *`	2	1002
`int *`, `float *`	4	1004
`double *`	8	1008
`int **` (64-bit)	8	1008
12-byte `struct *`	12	1012

Operation	OK?	Result
`p + n`, `p - n`, `p++`, `p--`	✓	moves n elements
`p2 - p1`	✓	elements between
`p1 < p2`, `p == NULL`	✓	compare positions
`p1 + p2`	✗	compile error
`p * 2`, `p / 2`, `p % 2`	✗	compile error
`void *` arithmetic	✗	not ISO C (gcc: 1 B)

```
int a[] = {10, 20, 30, 40};
int *p = a;
int x = *p++;   /* x=10, p->a[1] */
int y = (*p)++; /* y=20, a[1]=21 */
int z = *++p;   /* z=30, p->a[2] */
printf("%d %d %d %d", x, y, z, a[1]);
                /* 10 20 30 21   */
int *q = &a[3];
printf("%d", (int)(q - a)); /* 3 */
```

Expression	Changes	Value used
`*p++` = `*(p++)`	**p**, after	old *p
`*++p`	**p**, first	new *p
`(*p)++`	**value**, after	old value
`++*p`	**value**, first	new value

Walking an array with a pointer: `for (p = a; p < a + n; p++) sum += *p;` — comparing with `a + n` (one past the end) is legal, but reading `*(a + n)` is not. Pointers into **different** arrays must not be subtracted or ordered with `<`; `==` is always allowed. Print a pointer difference with `%td`.

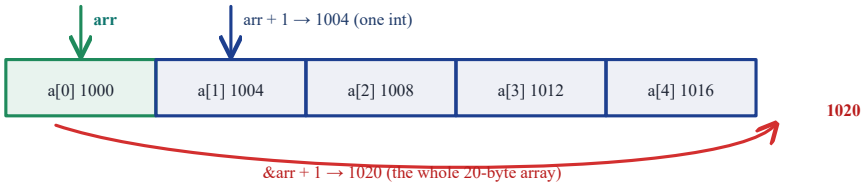
Memory trick: "**The type is the ruler.**" A pointer counts in elements; its type says how long one step is.

Exam trap: `*p++` moves the **pointer**, not the value: postfix `++` binds tighter than `*`. Two int pointers 12 bytes apart differ by **3**.

Worked example ✍ address: `double *d` holds 2000: `d + 5` = 2000 + 5×8 = **2040**. As `char *`: 2005.

5 1D Arrays and the Array–Pointer Link

Definition: An **array** is a fixed-size collection of elements of the **same type** in **contiguous** memory, indexed `0` … `n-1`; `int arr[5]` reserves 5 × 4 = 20 bytes. C does **no bounds checking**: `arr[5]` compiles and is undefined behaviour. In most expressions the name **decays** to a pointer to element 0, so **arr[i] ≡ *(arr + i)** ≡ `*(i + arr)` ≡ `i[arr]`.



arr and &arr print the same number; their types and steps differ

Expression	Type	Value
`arr`, `&arr[0]`	`int *`	1000
`arr + 1`	`int *`	1004
`&arr`	`int (*)[5]`	1000 (same!)
`&arr + 1`	`int (*)[5]`	1020
`sizeof(arr)`	`size_t`	**20**
`sizeof arr / sizeof arr[0]`	`size_t`	5

Declaration	Contents
`int a[5]={1,2};`	1 2 0 0 0
`int a[]={1,2,3};`	size taken as 3
`int a[5]={0};`	all zero
`int a[5];` local · global	garbage · all zero
`int a[5]={[2]=7};`	0 0 7 0 0 (C99)
`int a[n];`	VLA: C99; optional in C11

```
int sum(int a[], int n) { /* int *a */
    int s = 0;
    for (int i = 0; i < n; i++)
        s += a[i];       /* *(a + i) */
    return s;  /* sizeof(a) = 8 here */
}
int arr[5] = {1, 2, 3, 4, 5};
int n = sizeof arr / sizeof arr[0];
sum(arr, n);   /* passes &arr[0] */
```

When the name does NOT decay: ① under `sizeof` → whole array size; ② under `&` → pointer to the whole array; ③ a string literal initialising a `char` array. The name is **not a modifiable lvalue**: `arr++`, `arr = p` fail; `p++` is fine. A function gets only the address, so its changes to `a[i]` reach the caller.

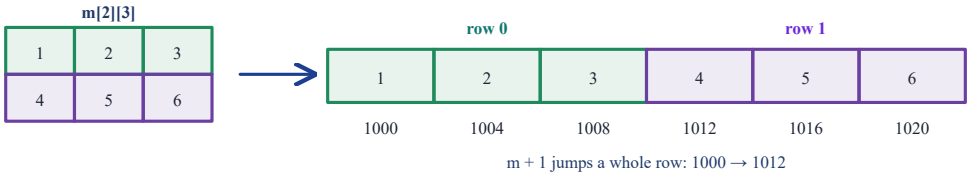
Memory trick: "**Array name = a fixed signpost to element 0; a pointer = a signpost you can carry.**"

Exam trap: An array parameter is a pointer: `sizeof(a)` inside the function gives 8 (or 4), not the array size. Always pass n.

Worked example ✍ output: `int a[5]={1,2,3,4,5}; int *p=(int *)(&a+1); printf("%d",*(p-1));` → **5**.

6 2D Arrays and the Address Formula

Definition: `int m[R][C]` is an array of R rows, each an array of C ints. Memory is one straight line, and C stores the elements in **row-major order**: all of row 0, then row 1, … Only the **first** (row) size may be left out — `int m[][3] = {…};`, `void f(int m[][3], int rows)` — because the **column size is required** to find where each row starts.



the grid is how we think; the line is how memory holds it

```
int m[2][3] = { {1, 2, 3},
                {4, 5, 6} };
for (int i = 0; i < 2; i++) {
    for (int j = 0; j < 3; j++)
        printf("%d ", m[i][j]);
    printf("\n");
}            /* 1 2 3
                4 5 6 */
int (*row)[3] = m; /* ptr to a row */
```

Expression	Means	Value
`m`	→ row 0	1000
`m + 1`	→ row 1 (+12 B)	1012
`*m + 1`	`&m[0][1]` (+4 B)	1004
`*(m + 1) + 2`	`&m[1][2]`	1020
`*(*(m + 1) + 2)`	`m[1][2]`	6
`sizeof m · m[0]`	all · one row	24 · 12

Address formula: M rows, N columns, W bytes per element, base address B, lower bounds L_r, L_c (both 0 in C).

Row-major: $LOC(A[i][j]) = B + W \times [N \times (i - L_r) + (j - L_c)]$

Column-major: $LOC(A[i][j]) = B + W \times [M \times (j - L_c) + (i - L_r)]$

C, C++, Pascal and NumPy (default) are row-major; Fortran, MATLAB and R are column-major.

Worked example ✍ A[1..10][1..15], W = 2, B = 100: find A[4][8]: M = 10, N = 15. **Row-major:** 100 + 2 × [15 × 3 + 7] = 100 + 2 × 52 = **204**. **Column-major:** 100 + 2 × [10 × 7 + 3] = 100 + 2 × 73 = **246**. In C, `int A[5][4]` at 1000: `&A[3][2]` = 1000 + 4 × (3×4 + 2) = 1056.

Worked example ✍ pointer to a row: `int a[2][3] = {{1,2,3},{4,5,6}}; int (*r)[3] = a;` → `r[1][1]` = 5, `*(*(r + 1))` = **4** (first element of row 1), `*(*r + 2)` = 3. `r + 1` moves 12 bytes because `*r` is a whole row of 3 ints.

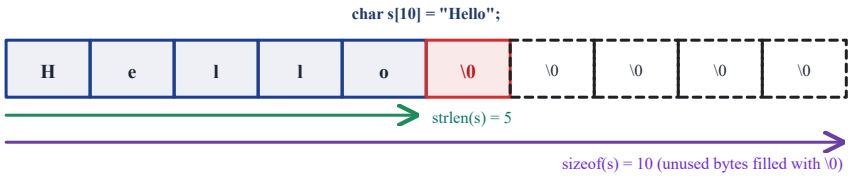
Memory trick: "**Row-major reads like a book**": finish a line, then the next. Column-major reads a newspaper column.

Exam trap: `int m[][] = {…};` is an error. `m + 1` jumps a row, `*m + 1` one element. Check whether bounds start at 0 or 1.

Worked example ✍ size: `int a[2][3][4]` holds 24 ints = **96 bytes**; `a[1]` is a 3×4 block of 48 bytes.

7 Strings and the Null Character

Definition: C has no string type: a **string** is a `char` array ending in the **null character** `'\0'` (ASCII 0). Every string function — `strlen`, `printf("%s")`, `strcpy` — walks until it meets `'\0'`. So the literal `"Hello"` occupies **6 bytes**: 5 letters + the terminator, added by the compiler.



strlen stops at the first '\0'; sizeof counts every byte the array owns

Declaration	Bytes	Note
`char s1[] = {'H','i','\0'};`	3	`'\0'` by hand
`char s2[] = "Hello";`	6	`'\0'` added
`char s3[10] = "Hello";`	10	rest `'\0'`
`char s4[5] = "Hello";`	5	**no `'\0'`**

Common misconception: The terminator is `'\0'`, value 0 — **not** `' '` (space, 32), `'0'` (digit, 48) or `'\n'` (10). `NULL` is a **pointer**; `'\0'` a **character**.

Point	`char s[] = "Dhaka";`	`char *p = "Dhaka";`
What it is	array: own copy, 6 bytes	pointer to a string literal
Change a letter	`s[0] = 'd';` ✓	`p[0] = 'd';` ✗ undefined (literal often read-only → crash)
Assign new text	`s = "Khulna";` ✗ → `strcpy`	`p = "Khulna";` ✓ re-points
`sizeof`	6	8 (pointer, 64-bit)

```
char s[] = "abc";
printf("%zu %zu", sizeof s, strlen(s)); /* 4 3 */
printf("%zu", sizeof("abc"));       /* 4 */
char *p = "Hello";
printf("%c %s", *(p + 1), p + 2);   /* e llo */
printf("%c", "Hello"[4]);           /* o */
char *names[] = {"Ali", "Rina"}; /* 2 ptrs */
```

`'A'` vs `"A"`: `'A'` is a character constant of type `int` in C: `sizeof('A')` = 4 (1 in C++). `"A"` is `{'A','\0'}`: `sizeof("A")` = 2. `char *names[]` = array of pointers; `char g[2][5]` = fixed 5-byte rows.

Memory trick: "**Every string carries an invisible full stop**": bytes needed = letters + 1.

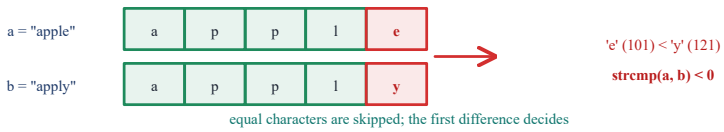
Exam trap: `s1 == s2` compares **addresses**, not text → `strcmp`. `char t[] = {'H','i'};` has no `'\0'`: `%s` runs on.

Worked example ✍ own strlen: `size_t len(const char *s){ const char *p = s; while (*p) p++; return p - s; }` → difference = **count**.

8 string.h Functions and String Input

Definition: `<string.h>` declares the functions for `'\0'`-terminated strings and raw memory blocks. None of them knows the destination's size: **the caller must leave room**, including the byte for `'\0'`. Comparison is **lexicographic by character code**.

Function	Does	Returns / note
`strlen(s)`	counts characters before `'\0'`	`size_t`; `'\0'` not counted
`strcpy(d, s)`	copies s with its `'\0'` into d	d; no size check
`strncpy(d, s, n)`	copies at most n chars	**no `'\0'`** if `strlen(s) ≥ n`
`strcat` · `strncat(d, s, n)`	appends s (≤ n chars) to d	d; `strncat` adds `'\0'`
`strcmp(a, b)` · `strncmp`	compares char by char	**0 if equal**, <0 or >0
`strchr` · `strrchr` · `strstr`	first c · last c · substring	pointer, or `NULL`
`strtok` · `memcpy` · `memset`	split · copy · fill raw bytes	`strtok` modifies s
`strrev`, `strupr`, `itoa`	reverse · upper case · int → text	**not standard C**



strcmp works like a dictionary, but by ASCII code

Input call	Stops at	Size check	Note
`scanf("%s", s)`	whitespace	no → `"%49s"`	no `&`: s is an address
`scanf("%[^\n]", s)`	newline	no	reads spaces too
`gets(s)`	newline	**none**	**removed in C11**
`fgets(s, n, stdin)`	newline (kept) or n−1	✓ safe	adds `'\0'`; `NULL` at EOF

```
char name[50];
fgets(name, sizeof name, stdin);
name[strcspn(name, "\n")] = '\0';
if (strcmp(name, "Rahim") == 0)
    puts("match");   /* puts adds \n */
char d[6];
strncpy(d, "Bangladesh", 5);
d[5] = '\0';         /* "Bangl" */
```

Why gets() went: No size parameter: a long line writes past the array over neighbouring memory — a **buffer overflow** (the 1988 Morris worm exploited one in `fingerd`). Deprecated in C99 (TC3), removed in C11. More I/O: later lesson.

Memory trick: "**strcmp = dictionary order by ASCII**": uppercase (65–90) sorts before lowercase (97–122).

Exam trap: `if (strcmp(a, b))` is true when the strings **differ**. Only the sign is guaranteed, not −1/+1.

Worked example ✍ mixed input: `scanf("%d",&n);` `fgets(s,50,stdin);` → fgets gets the leftover `'\n'`: an **empty line**.

Night-Before-the-Exam Sheet

Fact sheet:

`<h>`: system dirs · `"h"`: own dir, then system

Directives: no `;`, end at the line end

`#` stringize · `##` token paste

`p + n` = address + n × sizeof(*p)

`arr[i] ≡ *(arr+i) ≡ i[arr]`

Row-major: $B + W[N(i-L_r) + (j-L_c)]$

`"Hello"` = 6 bytes; `'\0'` = ASCII 0

Pipeline: preprocess → compile → assemble → link

Guard: `#ifndef X / #define X / #endif`

`__FILE__ __LINE__ __DATE__ __TIME__`

`p2 - p1` = elements, not bytes

No decay: `sizeof arr`, `&arr`

Column-major: $B + W[M(j-L_c) + (i-L_r)]$

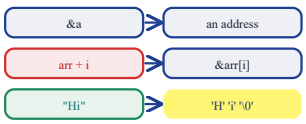
`strcmp`: 0 equal, <0, >0

Top 12 one-liners for MCQs

1. `printf/scanf` → stdio.h; `malloc` → stdlib.h
2. `conio.h` is not standard C
3. `SQUARE(2+3)` with `x*x` → 11
4. `const int *p`: value fixed; `int *const p`: pointer fixed
5. `void *` cannot be dereferenced without a cast
6. `*p++` moves p; `(*p)++` changes the value
7. Pointer size: 4 B (32-bit), 8 B (64-bit)
8. `&arr + 1` skips the whole array
9. Array parameter = pointer; pass the length
10. Only the first dimension may be omitted
11. `strlen("abc")` = 3, `sizeof("abc")` = 4
12. `gets` removed in C11 → `fgets`

Top traps: `#define` with `;` · `100/SQ(5)` = 100 · `int *x, y;` · dangling after `free` · `sizeof` of an array parameter · `m[][]` without columns · modifying `char *p = "..."` · `==` on strings · `strncpy` without `'\0'`.

All tricks together: shelf vs desk · brackets inside, brackets outside · & asks where, * asks what's there · the type is the ruler · fixed vs carried signpost · row-major reads like a book · invisible full stop · strcmp = dictionary by ASCII.



three ideas, three lines

Worked example ✍ one snippet, five facts: `char s[] = "Dhaka"; char *p = s + 1;` → `sizeof s` = 6, `strlen(s)` = 5, `*p` = 'h', `p[2]` = 'k', `p - s` = **1**. And `*s = 'd'` is fine because s is an array, not a literal.

✓ Practice MCQs (exam pattern, 10 marks)

1. With `#define SQUARE(x) x*x`, `SQUARE(3+1)` evaluates to:
(a) 16 (b) 8
(c) 7 (d) 10

2. Inside `void f(int a[])` on a 64-bit system, `sizeof(a)` is:
(a) 8 (b) 20
(c) 5 (d) 4

3. Which header declares `strlen()`?
(a) stdio.h (b) stdlib.h
(c) ctype.h (d) string.h

4. An `int *p` (4-byte int) holds 2000. `p + 3` is:
(a) 2003 (b) 2012
(c) 2006 (d) 2024

5. `const int *p;` means:
(a) `*p` is read-only (b) `p` is read-only
(c) both are fixed (d) neither is fixed

6. `int A[5][4]`, base 1000, 4-byte int, row-major. Address of `A[3][2]`:
(a) 1052 (b) 1048
(c) 1056 (d) 1060

7. Which function was removed from the C11 standard?
(a) fgets (b) gets
(c) puts (d) scanf

8. `char s[] = "Dhaka";` — `sizeof(s)` is:
(a) 5 (b) 4
(c) 8 (d) 6

9. `strcmp("abc", "abd")` returns:
(a) 0 (b) a positive value
(c) a negative value (d) 3

10. A header is kept from being included twice by:
(a) `#pragma pack` (b) `#ifndef` / `#define` / `#endif`
(c) `#error` (d) `#undef`

Answer key → 1-c · 2-a · 3-d · 4-b · 5-a · 6-c · 7-b · 8-d · 9-c · 10-b

Written question: Explain the relationship between arrays and pointers in C. Show how `arr[i]` is evaluated, why `sizeof` gives different results for an array and a pointer, derive the row-major address of `A[i][j]`, and write a pointer-based function that returns the length of a string. 10 marks

1. **Decay (2)** — an array name in an expression becomes `&arr[0]`; `arr[i]` is defined as `*(arr + i)`, and `arr + i` adds `i × sizeof(int)` bytes. 2
2. **sizeof (2)** — no decay under `sizeof`/`&`: `int arr[5]` → 20; a pointer (or array parameter) → 8 on 64-bit; the name cannot be assigned or incremented. 2
3. **Address (3)** — rows stored one after another: skip i full rows of N elements, then j: LOC = B + W[N(i − L_r) + (j − L_c)]; e.g. `int A[5][4]` at 1000 → `A[3][2]` = 1056. 3
4. **Function (3)** — `size_t len(const char *s){ const char *p = s; while (*p) p++; return p - s; }`: walk to `'\0'`, pointer difference = length. 3

Before the exam ✓

- ✓ Can name the header of any common function
- ✓ Can expand a macro and spot the trap
- ✓ Can compute `p + n` and `p2 - p1`
- ✓ Can tell `arr`, `&arr`, `sizeof` apart
- ✓ Can apply both 2D address formulas
- ✓ Can count string bytes with `'\0'`